J. Dairy Sci. TBC
https://doi.org/10.3168/jds.2026-29253
© TBC, The Authors. Published by Elsevier Inc. on behalf of the American Dairy Science Association®.
This is an open access article under the CC BY license (https://creativecommons.org/licenses/by/4.0/).

Influence of passive training of primiparous cows and its interaction with individual cow personality on adaptation to and performance on an automated milking system

J. E. Brasier,[1] R. Bergeron,[1] D. B. Haley,[2] and T. J. DeVries[1]*
[1]Department of Animal Biosciences, University of Guelph, Guelph, ON, N1G 2W1, Canada
[2]Department of Population Medicine, University of Guelph, Guelph, ON, N1G 2W1, Canada

ABSTRACT

The primary study objective was to determine how previous exposure to an automated milking system (AMS) pen via an exposure period to the AMS pen before first milking, giving cows the opportunity for previous exposure and social transmission, influences the adaptation of primiparous dairy cows to the AMS. A secondary objective was to determine whether certain personality types adapt better if they experienced an exposure period or not. Thirty-two primiparous Holstein dairy cows were enrolled as pairs in the study at 147 ± 6.97 DIM (mean ± SD) and were assessed for personality traits using a combined arena test comprised of 3 parts: a novel arena test, a novel object test, and a novel human test. Cows were allocated into 1 of 2 treatments within pair (n = 16 per treatment): control (CON; no exposure period), and exposure (EXP; had a 48-h exposure period in a free-flow AMS pen before first milking in the AMS, a form of passive training). During the exposure period, the EXP cows were kept in the AMS pen for 48 h, while being milked in the parlor 2 × /d, where they were exposed to the AMS pen, pen mates, were able to voluntarily explore the AMS, and were able to watch other adapted cows in the pen utilize the AMS. At the end of the 48-h period, both cows within the pair were milked in the AMS for the first time and remained in the AMS pen thereafter. Milking variables (visits, kickoffs, milk flow), behavior (activity, rumination, feeding), and yield were recorded for the exposure period (d −2 and −1) and the first 21 d in the AMS. Cows that experienced the exposure period had fewer meals (8.4 vs 12.1 meals/d; SE = 0.7) on d 0 (first day of milking in the AMS) compared with CON cows. From d 2 to 20 of milking in the AMS, the EXP cows had more frequent visits to the AMS (6.8 vs 5.5 visits/d; SE = 0.28), tended to have a 1.4 × lesser odds (odds ratio [OR] = 0.72; 95% CI = 0.51–1.01) of having a daily fetching event, and had greater milk yield compared with CON cows (33.5 vs 32.5 kg/d; SE = 0.29). The personality traits of boldness and activeness were identified in the cows. Personality was only associated with outcome variables for cows that experienced the exposure period, where the more bold EXP cows had lesser odds of experiencing a rejection (OR = 0.40; 95% CI = 0.26–0.59), voluntary visit (OR = 0.40; 95% CI = 0.26–0.59), and voluntary milking (OR = 0.34; 95% CI = 0.22–0.53), while more active EXP cows had greater odds of experiencing a voluntary visit (OR = 4.52; 95% CI = 1.22–16.81) compared with less bold and less active EXP cows. Overall, the results demonstrate that passive training, specifically an exposure period in the AMS pen before first milking, can improve adaptation to the AMS and milk letdown, and in turn milk yield. In addition, individual cow personality influenced how well those cows that experienced the exposure period adapted to the AMS.

Key words: social transmission, robotic milking, behavior, adaptation, personality

INTRODUCTION

The automated milking system (AMS) has the potential for greater milking frequency and yield, compared with milking 2 × /d on a conventional parlor, as well as improved cow welfare (Jacobs and Siegford, 2012a). However, if cows do not adapt to the AMS, measured by how often they voluntarily milk (ideally ≥ 3 × /d; Rodenburg, 2017), they need to be fetched or brought to the AMS by a human to be milked, resulting in a lesser milking frequency and lesser milk yield (Brasier et al., 2025a). Several researchers have demonstrated that previous exposure to the AMS, through habituation or training, can improve adaptation and performance when milked in the AMS (Jago and Kerrisk, 2011; Widegren, 2014; von Kuhlberg et al., 2021; Whitehead, 2024), resulting in

Received July 8, 2026.
Accepted September 13, 2026.
*Corresponding author: Trevor DeVries, Email: tdevries@uoguelph.ca

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

ARTICLE IN PRESS—UNCORRECTED PROOF

fewer fetching events and greater milk yield (Brasier et al., 2025a, b). Many of these studies were focused on the use of active training, which involves instructing cows to enter the AMS following a strict protocol. Only one study by von Kuhlberg et al. (2021), however, was focused on passive training, which Brasier et al. (2026) defined as cows being allowed to voluntarily explore or enter the AMS on their own before first milking in the AMS, and are not instructed by humans to enter the AMS following a strict training protocol. This passive training method by von Kuhlberg et al. (2021) resulted in improved adaptation across the first 5 d in the AMS. This type of training method may involve less time and labor during training compared with active training methods due to human assistance to enter the AMS not being involved in the training process.

In a previous study focused on actively training cows to the AMS, it was suggested that the cows that had no training adapted better, or equally well, to the AMS due to social transmission, compared with those that were trained, as demonstrated by no differences in visits to the AMS (Brasier et al., 2025a). Social transmission, the cognitive process involving the transfer of information from one animal to another by observing the consequences of their actions (Galef, 1988; Nawroth and Rørvang, 2022), aids cattle in learning to choose what to eat (Velázquez-Martínez et al., 2010; Costa et al., 2016), as well as avoiding aversive events (Munksgaard et al., 2001; Keshavarzi et al., 2020). The mechanisms of social transmission include social facilitation and stimulus enhancement. Social facilitation is where cattle performing a behavior can increase the motivation for others to also engage in that behavior (Rørvang et al., 2018), such as using feed bins and brushes (Nawroth and Rørvang, 2022; Stenfelt et al., 2022b). Stimulus or local enhancement, the other mechanism of social transmission, involves cows becoming motivated to obtain a stimulus or explore a location where a demonstrator cow is interacting with a motivating stimulus (Rørvang et al., 2018), such as the concentrate provided near the AMS exit (Wechsler and Lea, 2007; Nawroth and Rørvang, 2022; Brasier et al., 2025a).

The concept of social transmission differs from social learning, which is defined as acquiring a new behavior after observing a conspecific perform the behavior (Rørvang et al., 2018). Social learning has been discussed by several cattle researchers (Munksgaard et al., 2001; Velázquez-Martínez et al., 2010; Costa et al., 2016); however, it has not actually been demonstrated to occur in cattle, as observer cows do not perform a new behavior based on the demonstrator (Nawroth and Rørvang, 2022; Stenfelt et al., 2022a). The observer cow likely focuses their attention on the stimulus or location the demonstrator interacted with, so they can utilize this information from the demonstrator for individual learning (Stenfelt et al., 2022a). Therefore, it is likely that most researchers who discuss social learning in cattle are referring to social transmission. This includes Wechsler and Lea (2007), who suggested that social learning can aid in learning how to use equipment in their environment, which may include automated milk feeders for calves or AMS for cows (Nawroth and Rørvang, 2022), which likely relates to local enhancement (Rørvang et al., 2018).

The personality of an individual cow, specifically the trait of sociability, may affect the amount of influence social transmission has on a cow. Personality has been defined as individual differences in behavior that are consistent across time and situations (Finkemeier et al., 2018) and has been associated with how well a cow adapts to the AMS (Brasier et al., 2023; Schwanke et al., 2024a, b). Personality may also affect social learning (Nogues et al., 2023), the more cognitive complex form of social transmission. In previous research on AMS training, more complicated training methods (exposure to the robotic arm, noises, and concentrate) masked the influence of cow personality on adaptation to the AMS (Brasier et al., 2025a; 2026). However, in those studies, the individual personality of cows that were trained using simpler methods or were not trained at all was associated with how well those cows adapted to the AMS, unlike the cows that were trained with more complicated methods (Brasier et al., 2025a; 2026). Therefore, as passive training is a simple form of previous exposure compared with active training methods, how well a cow adapts to the AMS in response to passive training methods may be associated with their individual personality.

The primary objective of this study was to determine how passive training through a 48-h exposure period in a free-flow AMS pen before first milking in the AMS, giving cows the opportunity to experience social transmission and previous exposure to the AMS (as per Brasier et al., 2025a), influences the adaptation of primiparous dairy cows to the AMS. A secondary objective was to determine whether cows of certain personality types adapt better depending on whether they experienced an exposure period or not. It was predicted that cows who experience an exposure period would adapt better to the AMS compared with cows who did not experience the exposure period. Further, it was predicted that cows with certain personality types, such as sociability, may be more effective at utilizing information received from conspecifics through social transmission.

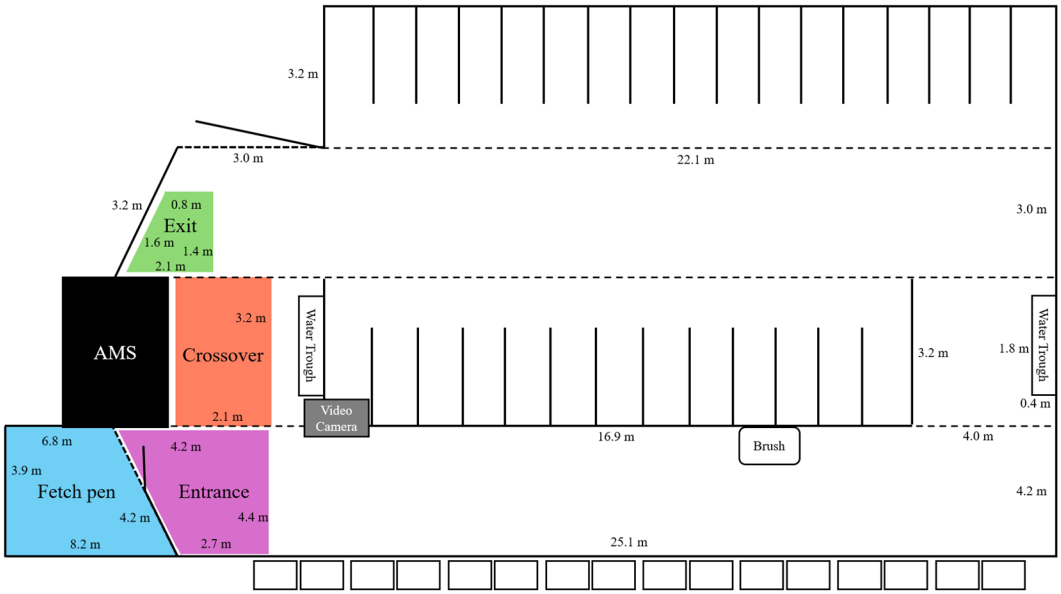


Figure 1. Pen layout and visual aid for ethogram (Table 3) of the observed behaviors from video recordings. Areas in color were within video camera view. The blue area is the fetch pen, the purple area is the entrance, the orange area is the crossover, and the green area is the exit of the AMS. The video camera was not able to capture any other areas of the pen shown in the diagram.

MATERIALS AND METHODS

Animals, Housing, and Feeding

Thirty-two primiparous Holstein cows were enrolled in this study between 140 and 160 DIM (DIM = 147 ± 6.97; mean ± SD) between March and August 2025. The study took place at the University of Guelph, Elora Research Station – Ontario Dairy Research Centre (Elora, ON, Canada). Any enrolled cow that would have developed severe clinical illness, such as lameness or mastitis, as determined by barn staff or the vet (as appropriate) would be examined, treated and removed from the study at any time point and replaced if needed. However, no cows were removed or replaced in the current study.

Throughout the study, animals were housed in 1 of 3 lactating pens, or in 1 AMS pen (Figure 1), all located on the same side of the barn. All pens had a maximum of 30 cows; within each pen they had access to 1 cow brush, 30 lying stalls, 2 water troughs at either side of the pen, and 16 automated feed bins (Hokofarm Group. Marknesse, the Netherlands). The free stalls (length of 295 cm; width of 126 cm; neck rail height of 125 cm above the stall base) in all pens were laid out in 2 rows (13 on one side 17 on the other), tail-to-tail. The stalls had a mattress base (Pasture Mat; ProMat, Woodstock, ON, Canada) and were bedded with chopped straw that was added once weekly, and were cleaned 2 × /d. Before their first AMS milking, cows were milked on the rotary parlor (Delaval, Tumba, Sweden) 2 × /d at 0400 and 1600 h. Within the single AMS pen, cows had access to 1 free-traffic AMS (DeLaval Voluntary Milking System, Tumba, Sweden), with a fetch pen adjacent to the entrance of the AMS (Figure 1).

The lactating cow total mixed ration (TMR) and partial mixed ration (PMR) for the singular AMS pen were prepared daily in a mixer wagon (Jaylor Model 5572, Jaylor Fabricating, Orton, ON, Canada) and fed once daily at 0900 h. The targeted refusal rate was 5–7% (as fed basis), which was adjusted based on the average of the previous 3 d. Feed bins were cleaned out daily before fresh feed being added. To achieve the specified DM provision of the AMS concentrate, the maximum amount offered in the AMS was 5.4 kg/d DM, starting at 2.5 kg/d DM for their first day in the AMS, and increased based on linear accrual over time (achieved on d 12 of milking in the AMS). For each milking, the concentrate allocation had a minimum allocation of 50 g per milking and a maximum of 2.50 kg. Ingredient and chemical composition of the TMR, PMR, and AMS concentrate pellet are shown in Table 1.

Personality Assessment. Cows were assessed for personality traits using a combined arena test (as described by Brasier et al., 2023) on the day of enrollment between 1100 and 1300 h. The observational arena (5 × 8 m) consisted of rubber mat flooring and metal fencing, where cows had access to a water trough and limited visual contact with conspecifics. The floor of the arena was marked into quadrants using chalk, along with 2 square boxes representing 1 m and 2 m from the center. The test started when the gate was closed after a familiar person brought

ARTICLE IN PRESS—UNCORRECTED PROOF

Table 1. Ingredient and chemical composition (mean ± SD) of the formulated total milked ration (TMR), partial mixed ration (PMR), and automated milking system (AMS) concentrate

Component	TMR	PMR	AMS Concentrate
Ingredients (% of DM)			
Corn silage[1]	39.1	45.8	—
Alfalfa haylage[2]	26.1	30.5	—
Wheat straw[3]	3.5	4.1	—
High moisture corn[4]	16.7	6.5	—
TMR supplement[5]	14.6	—	—
PMR supplement[6]	—	13.1	—
AMS concentrate[7]	—	—	100
Chemical composition[8]			
DM (%)	43.5 ± 1.24	40.5 ± 1.11	90.7 ± 0.37
CP (% of DM)	14.5 ± 0.63	14.5 ± 0.75	19.0 ± 0.36
ADF (% of DM)	21.9 ± 1.13	25.4 ± 0.52	14.0 ± 0.32
NDF (% of DM)	30.7 ± 0.50	35.3 ± 0.96	32.3 ± 0.57
Starch (% of DM)	24.1 ± 1.36	19.0 ± 0.48	20.1 ± 0.70
NE_L (Mcal/kg DM)	1.6 ± 0.02	1.6 ± 0.01	1.8 ± 0.01
Fat (% of DM)	4.9 ± 0.51	4.9 ± 0.12	3.2 ± 0.39
Ash (% of DM)	6.6 ± 0.31	7.0 ± 0.14	7.2 ± 0.29
Ca (% of DM)	0.8 ± 0.05	0.8 ± 0.02	1.3 ± 0.15
P (% of DM)	0.3 ± 0.02	0.3 ± 0.02	0.7 ± 0.01

[1]Corn silage had a DM of 34.6% and chemical composition (DM basis) of 6.9 ± 0.36% CP, 33.2 ± 2.73% starch, 21.9 ± 2.04% ADF, and 36.7 ± 1.72% NDF.

[2]Alfalfa haylage had a DM of 36.8% and chemical composition (DM basis) of 16.9 ± 2.74% CP, 1.98 ± 0.47% starch, 34.3 ± 3.53% ADF, and 45.4 ± 5.77% NDF.

[3]Wheat straw had a DM of 91.0% and chemical composition (DM basis) of 5.6 ± 0.69% CP, 3.2 ± 0.18% starch, 52.5 ± 2.09% ADF, and 78.0 ± 2.06% NDF.

[4]High moisture corn had a DM of 71.4% and chemical composition (DM basis) of 8.4 ± 0.59% CP, 63.5 ± 4.05% starch, 2.4 ± 0.49% ADF, and 6.9 ± 0.52% NDF.

[5]Supplied by Floradale Feed Mill Ltd. (Floradale, Ontario, Canada) including ingredients (as-fed); 32.5% high-bypass soybean meal, 34.5% soybean meal 47%, 5.0% canola, 5.0% Hi pro corn gluten (60%; Hi-Pro Feeds, Texas, USA), 4.8% Palmit 80 (Global Agri-Trade Co., California, USA), 2.7% sodium sesquicarbonate, 3.1% fine salt, 1.6% magnesium oxide, 3.7% limestone calcium carbonate, 1.4% Urea, 1.4% monocalcium phosphate (Dical), 2.0% tallow (animal/vegetable blend), 0.6% organic ruminant micronutrient premix (Floradale Feed Mill Ltd.), 0.6% granular sulfur (99.5%), 0.33% yeast culture (Diamond V Yeast XP Diamond V, Cedar Rapids, IA), 0.07% zinc amino acid complex (Availa Zn 120, Zinpro, Eden Prairie, MN), 0.05% Rumensin/Coban (Elanco, Greenfield, IN), 0.03% organic selenium, 0.02% vitamin E (200KIU/kg), 0.6% methionine (Alimet, Novus International, Inc., Chesterfield, MO), and 0.02% biotin supplement (Rovimix Biotin 20000, DSM Animal Nutrition and Health, Maastricht, the Netherlands).

[6]Supplied by Floradale Feed Mill Ltd. (Floradale, Ontario, Canada) including ingredients (as-fed); 38.0% high-bypass soybean meal, 9.6% soy hulls (ground), 20.0% soybean meal 47%, 5.0% Hi pro corn gluten (60%; Hi-Pro Feeds, Texas, USA), 5.0% Palmit 80 (Global Agri-Trade Co., California, USA), 5.0% canola, 3.1% sodium sesquicarbonate, 3.5% fine salt, 1.6% magnesium oxide, 3.2% limestone calcium carbonate, 0.9% monocalcium phosphate (Dical), 1.6% Urea, 1.0% tallow (animal/vegetable blend), 0.6% organic ruminant micronutrient premix (Floradale Feed Mill Ltd.), 0.6% granular sulfur (99.5%), 0.44% yeast culture (Diamond V Yeast XP Diamond V, Cedar Rapids, IA), 0.09% zinc amino acid complex (Availa Zn 120, Zinpro, Eden Prairie, MN), 0.06% Rumensin/Coban (Elanco, Greenfield, IN), 0.04% organic selenium, 0.01% vitamin E (200KIU/kg), 0.6% methionine (Alimet, Novus International, Inc., Chesterfield, MO), and 0.03% biotin supplement (Rovimix Biotin 20000, DSM Animal Nutrition and Health, Maastricht, the Netherlands).

[7]Supplied by Floradale Feed Mill Ltd. (Floradale, Ontario, Canada) including a proprietary mix of wheat shorts, soybean meal, ground soy hulls, beet pulp, ground barley, ground wheat, bakery meal, ground corn, molasses, limestone calcium carbonate, pellet binder and die lubricant (Pelltech; Borregaard LignoTech, Sarpsborg, Norway), fine salt, magnesium oxide, FFM Ruminant Micronutrient Premix (Floradale Feed Mill Ltd.), and flavor (Covotek Fenugreek 232; Allied Nutrition, Centurion, South Africa).

[8]Values were obtained from chemical analysis of TMR, PMR, and concentrate samples.

the cow to the arena. The test consisted of 3 consecutive parts: 1) a 10-min novel arena test; followed by: 2) a 10-min novel object test where an unknown object (a plastic chair (44.5 × 49 × 76 cm)) was placed in the center of the area; directly followed by: 3) a 10-min novel human test, where an unknown human stood at the center of the arena in clothing that was novel to the cows (bright blue coveralls with high visibility banding). Any cow suspected or confirmed in heat was not tested on that day, but rather at least 3 d later.

Each 30-min arena test session was video recorded using 2 cameras (GoPro Max, GoPro Inc., San Mateo, CA, USA and Handycam Camcorder, Sony Corporation, Tokyo, Japan; see Schwanke et al. (2024a) for a diagram

showing where the cameras were placed) to conduct continuous focal behavior sampling for each stage of the combined arena test (novel arena, novel object, and novel human) using the ethogram described in Table 2. Video analysis was performed by a single researcher after establishing interobserver reliability of ICC >0.80 for each test between 2 observers, and intraobserver reliability of ICC >0.93 using 20% of the combined area test videos (7 cows). Training for behavior scoring was provided through a detailed explanation of the ethogram. All time duration variables were converted to a percentage of test time to account for time excluded from analyses due to time spent being distracted (e.g., by a human walking by or moving other cows, or a vehicle driving by) or not in view, calculated as time spent performing the behavior, divided by total test time. If applicable, the time the cow was not visible during the assessment was subtracted from the total assessment time before calculation (a max of 19 s per 10 min).

Experimental Design

Cows of similar DIM were blocked and enrolled in pairs that were followed together for the entire 23-d period. Within pairs, cows were allocated to 1 of 2 treatments, where treatments were balanced for DIM, average milk yield in the parlor for the last 7 d before enrollment (34.1 ± 4.53 kg/d; mean ± SD), and latency to contact the novel object and human from the personality test: 1) cows in the control treatment (**CON**; n = 16) were not exposed to the AMS until first milking, and 2) cows in the exposure treatment (**EXP**; n = 16) were subjected to a 48-h exposure period where they were kept in the AMS pen, but not milked in the AMS, before their first AMS milking. The exposure period involved experience with the AMS pen and pen mates, and also gave EXP cows the opportunity to voluntarily explore the AMS and watch other adapted cows utilize the AMS. This gave EXP cows the opportunity to use social transmission while learning about the AMS and the associated environment during this time. Sample size and power analyses were used to calculate (Morris, 1999) the minimum number of replicates needed per treatment (n = 16) to detect a 35% reduction in the frequency of fetching events to the AMS. Estimates of variation (average CV = 47.5%) for the variable of fetching events were based on previously reported values (Schwanke et al., 2022; 2024b; Brasier et al., 2023).

Exposure Period. After personality testing, EXP cows (one from each pair) were moved into the free traffic AMS pen after the evening milking in the parlor (d −2 relative to the first day of milking in the AMS; d started at 1600 h) on the same day. The EXP cows were kept in the AMS pen and milked in the parlor at normal milking times for a total of 48 h to have the opportunity to experience social transmission from the other cows in the AMS pen (average number of cows in the pen = 29.3 ± 1.6; average parity of cows in the pen = 1.7 ± 0.3; average DIM of cows in the pen = 193.4 ± 18.7), who were all adapted and milked in the AMS for at least a month before the study began, as well as cows enrolled in the current study that were added across the study period. A period of 48 h was chosen to account for the effects of regrouping. While regrouping effects can last up to 2 d following the day of regrouping (von Keyserlingk et al., 2008), it typically mainly influences cow behavior during the first 24 h in a new pen (Marumo et al., 2024). Thus, the 48-h period was chosen to allow for a sufficient amount for time (i.e., 24 h) for cows to adjust to the new pen and an extra 24 h to potentially experience social transmission and habituation to the external environment of the AMS, without experiencing the majority of the stress they may

Table 2. Ethogram used for scoring personality behavior videos.[1]

Behavior	Definition
Recorded for novel arena, object and human test	
Vocalizations (count)*	All types of vocalizations made by the cow
Vigilance (sec)	Cow raises their head above the withers while stationary
Attraction (sec)	Cow raises their head above the withers while stationary looking in the direction of the home pen
Locomotion (sec)	Cow moves all four legs to cover distance
Exploratory (sec)	Cow sniffs or licks an object or surface in the testing arena
Escape (sec)	Cow head is through or over the railing of the testing arena
Changing quadrants (count)	When a cow switches to a new quadrant
Recorded for novel object and human test only	
2m away, within 1m, and contact with novel object/human (sec)	Any part of the cow's body in physical contact with the novel object or person
Latency to contact novel object/human (sec)	Time from the start of the test to first contact with the novel object or human.
Time spent behind novel person (sec)	Time spent in the two quadrants behind the novel person's back
Attention (sec)	Cow focuses their attention to the object or human

*Separate events were 10 s between each count. .

[1]Adapted from Brasier et al. (2023).

experience that is related to regrouping. There is limited information on the ideal time for cows to experience the transfer of information, with some studies having short observational sessions (30 min max; Bailey et al., 2000; Stenfelt et al., 2022a), and others grouping inexperienced with experienced cows where behavior was followed from that point onward (Ralphs et al., 1994; Costa et al., 2016). Therefore, 48 h was chosen to give cows time to adjust to regrouping, give them ample opportunity to explore the AMS and observe other cows, as well as account for changes in behavior due to circadian rhythm. For milking during that 48-h time period, EXP cows were sorted out of the AMS pen, put in the parlor before any other cow in the herd, then sorted back into the AMS pen immediately after exiting the parlor. The AMS was not able to prevent EXP cows from entering in the AMS during this period. However, the AMS was programmed to not milk EXP cows if they entered and open the exit door immediately if they did enter so that they would not be milked by the AMS before the first milking. The CON cow paired with the EXP cow remained in their respective lactating pen away from the AMS during this time and was milked in the parlor at normal times with their pen mates.

First Milking and Adaptation. Forty-eight h after EXP cows were moved to the AMS pen (i.e., at the end of d −1), the CON cow from the pair was moved into the AMS pen at approximately 1600 h on d 0 and milked for the first time in the AMS, without having been milked in the parlor that evening. The EXP cow from the pair was also not milked in the parlor that evening, and also experienced their first AMS milking at approximately 1600 h. The treatment that was milked first within pair was alternated for each pair. All cows remained in the AMS pen after the first milking.

When the cow entered the AMS for the first time, researchers adjusted the feed trough to fit the cow, and then the teat coordinates were saved automatically by the AMS, while the researcher observing the milking stopped or stepped in when needed, such as a cow kicking at the arm or milker. Milking permission in the AMS was granted when 6 h had elapsed since the previous milking and predicted milk yield for that milking exceeded 10.0 kg. The cows had access to the AMS for 22.5 h/d, with the remaining time dedicated to 30-min equipment cleaning cycles at 0730, 1530, and 2330 h daily. All cows were then brought by the researchers to the AMS to be milked from within the AMS pen for the remainder of d 0 at 0800 and 1400 h, then on d 1 at 2000, 0800, and 1400 h, as well as at 2000 h on d 2 for a total of 7 milkings observed by researchers. Cows were set to pass through the AMS if they entered in between the milking times for the first 7 milkings so that all cows were milked the same number of times by researchers. After, for the remainder of days in the AMS (d 2 to 20), cows were fetched to be milked by barn staff 2 × /d at 0400 and 1600 h, when notified by the AMS at those times if cows had not been milked in 10 h. For fetching, cows were placed in the fetch pen by the entrance to the AMS (see Figure 1), where they needed to pass through the AMS to gain access to stalls and feed. All fetching activity was recorded by barn staff on data sheets provided, as well as confirmed from video recordings obtained via security camera (Reolink RLC-511W; City of Industry, CA, USA; see Figure 1).

Milking, Behavior, and Yield Data

Scoring for ease of entry to the AMS, kicking, AMS concentrate consumption, and milk letdown were done for the first 7 milkings by 2 observers. All scoring criteria were from Brasier et al. (2025b), where a higher score is worse. Ease of entry score was scored from 0 to 6, where interobserver reliability was $\kappa > 0.84$ between 2 observers, and intraobserver reliability was $\kappa > 0.90$. Scoring was assigned as follows: cow enters on her own immediately (0), had to be directed toward the AMS entrance by a human without touching (1), had to be lightly touched by a human to direct them into the AMS (2), had to be pushed in by a human (3), had to use a gate (4), had to use a gate that was pushed for more than 15 s (5), and had to use a gate and extra people to enter (6). Kicking was scored on a 4-point scale from 0 to 3, where interterobserver reliability was $\kappa > 0.82$ between 2 observers, and intraobserver reliability was $\kappa > 0.88$. Scoring was assigned as follows: no kicking or stepping (0), only stepping (1), ≤ 2 kicking events and stepping if applicable (2), > 2 kicking events and stepping if applicable (3). Concentrate (pellet) consumption in the AMS was scored on a 3-point scale from 0 to 3, where interobserver reliability was $\kappa > 0.80$ between 2 observers, and intraobserver reliability was $\kappa > 0.88$. Scoring was assigned as follows: ate concentrate offered immediately (head was in the feed trough for majority of the session = 0), ate some concentrate offered (head was in the feed trough for some of the session, some concentrate left over = 1), did not touch the concentrate offered (head did not go into the feed trough = 2). Milk letdown was scored on a 4-point scale from 0 to 3, where interobserver reliability was $\kappa > 0.86$ between 2 observers, and intraobserver reliability was $\kappa > 0.94$. Scoring was assigned as follows: bad (teat cups were removed by AMS multiple times [>2] due to low flow; at less than 300 g/min; 3), poor (very slow to start, some teat cups [≤2] were removed by AMS due to low flow of less than 300 g/min; 2), acceptable (low milk letdown at first, at less than 300 g/min, but no teat cups removed; 1), and good (normal flow, no teat cups removed; 0). Scoring was done at the same time as milking, where the researcher was not blind to treat-

ment, as they were the ones responsible for moving the cows into the AMS pen. However, the second observer, who scored based on videos at a later time, was blind to treatment.

AMS visits, milkings, rejections, kickoffs, milk flow rate, box time duration, milking duration, attachment duration (time to clean and attach teat cups), and yield per milking were measured throughout the study by the AMS system (DelPro 4.5, DeLaval, Tumba, Sweden) and were used to determine the daily frequency of each variable. The camera at the AMS was used to identify which milkings for each cow were voluntary, as well as the score for ease of entry for fetching events by barn staff from d 2–20, using the same score described previously. The video recordings were also used to determine the first day each cow achieved a voluntary visit to the AMS, as well as the first day each cow achieved a voluntary milking in the AMS relative to their first day of milking in the AMS. Also, time spent around the AMS and social behaviors displayed around the AMS (Figure 1) were observed from the video recordings (the behaviors of time spent at the entrance, crossover, exit, and within the fetch pen were adapted from definitions provided in Brasier et al. (2025b); the social behaviors of displacements at the AMS entrance, allogrooming, pushing, head butting, and fighting were recorded when the cow of interest was both the actor and receiver of the behavior; see ethogram in Table 3). The behaviors observed using the video recordings were recorded using continuous focal animal sampling (Martin and Bateson, 2021) during the 48-h exposure period for EXP cows (d −2 and −1), and on d 0, 2, 6, 13, and 20 in the AMS for every cow for each 24-h period (from 1600 to 1559 h the next day). The 2 exposure days were chosen to observe if cows were near the AMS for them to potentially experience social transmission, and the 5 d during the adaptation period were based on a previous study that also observed cows on d 0, 2, 6, 13, and 20 (Brasier et al., 2025b). Time in the fetch pen was observed from d 2 to 20 instead of the 5 d, as fetching only occurred 2 × /d at specified times. Cows were identified on video recordings using haircoat pattern, as well as chalk letters that were drawn on their back on observation days. Interobserver reliability of behavior scoring from video met the ICC >0.80 cutoff between 3 observers after observing multiple videos of specific behaviors outlined in the ethogram, as well as 10 d of videos (24 h each), where 2 observers were blinded to treatment. Specific ICC values for each of the recorded behaviors are presented in Table 3.

An electronic monitoring (HR-TAG-LD, SCR Engineers Ltd., Netanya, Israel), attached to a nylon collar on each cow, was used to monitor rumination and activity (Merenda et al., 2020), stored at 2-h intervals. These data were used to determine total daily rumination (min/d) and activity time for each cow (arbitrary units; u/d) from d −2 to 20.

Automated feed bins (Hokofarm Group. Marknesse, the Netherlands) were used to monitor feeding behavior and intake from each feed bin visit for each individual cow (as validated by Chapinal et al., 2007). These data were used to calculate daily intake (kd/d), feeding time (min/d), and feeding rate (kg/min). Meal criteria were also calculated for each cow using the data collected from the feed bins, or the minimum time interval between meals of each cow, using a software package (MIX 3.1.3; MacDonald and Green, 1988) to fit normal distributions to the frequency of $\log_{10}$ transformed time intervals between bin visits (as per DeVries et al., 2003). These meal criteria were then used to determine meal frequency (no./d), meal duration (min/meal), and meal size (kg/meal) for each cow.

Feed Sampling and Analyses

Feed samples of the freshly mixed feed were taken once monthly for the 2 diets (TMR and PMR) and also for all ration ingredients (including the AMS pellet). Feed samples were immediately oven-dried at 60°C for 48 h for DM analysis after collection. After being dried, the feed samples were ground through a 1-mm sieve (Model 4 Wiley Laboratory Mill, Thomas Scientific, Swedesboro, NJ) and then shipped to A&L Canada Laboratories Inc. (London, ON, Canada) for analysis of CP (N × 6.25; AOAC International, 2000, method 990.03; Leco FP-628 Nitrogen Analyzer, Leco Corp., St. Joseph, MI), ADF (AOAC International, 2000, method 973.18), NDF with heat-stable α-amylase and sodium sulfite (AOAC International, 2000, method 2002.04), starch (heat-stable amylase and amyloglucosidase; AOAC International, 2000, method 996.11), crude fat (using petroleum ether; AOAC International, 2000, method 920.39), ash (550°C; AOAC International, 2000, method 942.05), and minerals (calcium and phosphorus; using aquaregia digestion inductively coupled plasma atomic emission spectroscopy).

Statistical Analysis

All statistical analyses were performed using the SAS 9.4 software (SAS Institute Inc., 2013). Significance was declared if $P \leq 0.05$ and tendencies were considered at $0.05 < P \leq 0.10$. For all data, days started at 1600 h and ended at 1559 h the next day, in accordance with the timing of the first milking in the AMS. All data were assessed for normality both visually and using the Shapiro-Wilk statistic (with $P > 0.05$ indicating normality) with the UNIVARIATE procedure of SAS. The personality behaviors of quadrant switches and vocalizations were

Table 3. Ethogram of the observed behaviors from the AMS security camera

Behavior	Definition	Reliability[1]
Entrance[2]	Time spent (min) in the entrance area of the AMS (purple area of Figure 1. Must have 2 front feet and head in area shown and be facing the general direction of the AMS. Max distance is 3.5 m of the AMS entrance gate). From Brasier et al. (2025b).	ICC >0.99
Exit[2]	Time spent (min) in the exit area of the AMS (green area of Figure 1. Must have 2 front feet and head in area shown and be facing the AMS exit. Max distance is 3 m of exit gate). From Brasier et al. (2025b).	ICC >0.99
Crossover[2]	Time spent (min) in the crossover watching the AMS (orange area of Figure 1. Must have 2 front feet and head in area shown and be facing the AMS. Max distance 2 m of the side gate). From Brasier et al. (2025b).	ICC >0.82
Fetch Pen[2]	Time spent (min) in the fetch pen (blue area of Figure 1. Must be in the fetch pen, time is from when the gate is shut by barn staff until the cow enters the AMS). From Brasier et al. (2025b).	ICC >0.99
Total time around the AMS[2]	Time spent (min) in the entrance, exit, and crossover area of the AMS. From Stewart et al. (2025).	ICC >0.93
Displacements[3,4]	Pushed or was pushed by another cow out of the way of AMS entrance, while standing ready to go in. Adapted from Foris et al. (2019).	Actor ICC >0.81 Receiver ICC >0.90
Allo Grooming[3,4]	Licking another cow or being licked by another cow anywhere on the body. Adapted from Rousing and Wemelsfelder (2006) and Foris et al. (2019).	Actor ICC >0.84 Receiver ICC >0.83
Pushing[3,4]	Pushing or being pushed by another cow by the body (body to body contact). Adapted from Rousing and Wemelsfelder (2006) and Ren et al. (2021).	Actor ICC >0.86 Receiver ICC >0.87
Head butting[3,4]	Hitting another cow with their head or being hit by another cow by the head (head to body contact). Adapted from Rousing and Wemelsfelder (2006) and Ren et al. (2021).	Actor ICC >0.89 Receiver ICC >0.91
Fighting[3]	Head-to-head contact, where both cows are pushing with their forehead. Adapted from Rousing and Wemelsfelder (2006) and Ren et al. (2021).	ICC >0.93

[1]Interobserver reliability using the intraclass correlation coefficient (ICC) of behavior scoring between 3 observers, where 2 observers were blinded to treatment.

[2]Duration (min/d).

[3]Count (#/d).

[4]Was recorded if cow of interest was the actor or receiver of the behavior (Rousing and Wemelsfelder, 2006).

transformed using the square root function to achieve normality. Also, the performance data of time in the fetch pen, time spent at the crossover, entrance, and exit of the AMS, the social behaviors of pushing, head butting, and fighting, and the personality behaviors of exploring, time in contact, time spent at 1 m, and latency to contact were transformed using the natural log function to achieve normality. Given non-normal distribution, the milking variables of fetching events, kickoffs, voluntary visits, voluntary milking, and rejections, as well as the social behaviors of displacements and grooming were categorized to binary variables (0 = no event and 1 = event) for each of the 21 d. Visits to the AMS were also non-normal, fitting a Poisson distribution, therefore AMS visits were analyzed with a Poisson distribution. Also, all scoring data during the first 7 milkings to the AMS (ease of entry score, kicking score, milk letdown score, and the score for eating concentrate in the AMS) were analyzed with a Poisson distribution given their non-normal distribution. The assumptions of normality were met for all other data.

Personality Testing Analysis. Behaviors scored across each of the 3 tests from the combined arena test (Table 2) were tested for an association across the 3 tests using the CORR procedure of SAS. Vocalizations (arena, object, and human), escaping (arena, object, and human), and quadrant changes (arena, object, and human) were significantly associated ($P \leq 0.05$) across tests and therefore were averaged across the 3 tests. A principal components analysis (PCA) was completed to identify personality traits using behaviors from the combined arena test. Following the procedure outlined by Budaev (2010) for animal behavior PCAs, the FACTOR procedure of SAS with varimax rotation was used for the PCA. First, behaviors with communality estimates < 0.30 were removed. Once the PCA had a sampling adequacy of > 0.50, which suggests an adequate level of correlations to obtain a valid result from the PCA (Budaev, (2010), factors with eigenvalues greater than 1.0 were further explored; 2 factors were retained in the current analysis. Behaviors that were retained in the PCA included quadrant switches during all 3 tests, locomotion in the object test, escaping across all 3 tests, exploring during the human test, time spent in contact with the novel human, and time spent at 1 m and 2 m from the novel human. Behaviors that loaded highly

ARTICLE IN PRESS—UNCORRECTED PROOF

on a factor from the PCA were considered equal to or greater than ± 0.63. These high loading behaviors were then used to interpret a personality trait from the factor. The score for each cow, ranging from highly negative to highly positive, was extracted for each factor.

Effect of the exposure period on behavior. Behaviors during the exposure period (d −2 and −1), including time spent around the AMS, social behaviors around the AMS, and feeding behaviors, were not analyzed due to differences in location, diet, group composition, and stocking density between the 2 treatments at that time. However, the effects of treatment on rumination, activity, and milk yield were analyzed using the MIXED procedure of SAS. The model included the fixed effects of treatment and exposure period day, and the random effect of pair. Day was the repeated measure, with cow within pair as the subject, and the variance-covariance matrix structure used was autoregressive or compound symmetry, chosen on the basis of best fit according to Schwarz's Bayesian information criterion. Degrees of freedom for fixed effects were estimated using the Kenward-Roger option in the MODEL statement. When an interaction of treatment and day was detected ($P \leq 0.05$), the data were analyzed by day. Several covariates were tested and retained in the model when significant ($P \leq 0.05$); these included: DIM at first AMS milking (retained for outcome: none), the average milk yield for the last 7 d in the parlor before enrollment and the exposure period (retained for outcome: yield), and the average rumination time for the last 7 d before enrollment and the exposure period (retained for outcome: rumination).

The effects of treatment on ease of entry score, kicking score, milk letdown score, and the score for eating concentrate in the AMS for the first 7 milkings in the AMS were analyzed using the GLIMMIX procedure of SAS, with a Poisson distribution. The model included the fixed effects of treatment and milking session (1–7), and the random effect of pair. Milking session (1–7) was the repeated measure, with cow within pair as the subject, and the variance-covariance matrix structure used was autoregressive or compound symmetry, chosen on the basis of best fit according to the generalized chi-squared to degrees of freedom ratio. Degrees of freedom for fixed effects were estimated using the Kenward-Roger option in the MODEL statement. When an interaction of treatment and milking session was detected ($P \leq 0.05$), the data were analyzed by session. Factor score from the PCA was also tested as a potential predictor in the model and retained when significant. An interaction of treatment and factor score was also tested; if an interaction of treatment and factor score was detected ($P \leq 0.05$), the data were analyzed by factor score within treatment.

The effects of treatment on milk yield, box time, milking duration, attachment duration, mean flow rate, and peak flow rate for the first 7 milkings in the AMS were analyzed using the MIXED procedure of SAS. The model included the fixed effects of treatment and milking session (1–7), and the random effect of pair. Milking session (1–7) was the repeated measure, with cow within pair as the subject, and the variance-covariance matrix structure used was autoregressive or compound symmetry, chosen on the basis of best fit according to Schwarz's Bayesian information criterion. Degrees of freedom for fixed effects were estimated using the Kenward-Roger option in the MODEL statement. When an interaction of treatment and milking session was detected ($P \leq 0.05$), the data were analyzed by session. Several covariates were tested and retained in the model when significant ($P \leq 0.05$); these included: DIM at first AMS milking (retained for outcome: none), the average milk yield for the last 7 d in the parlor before enrollment and the exposure period (retained for outcome: yield), the average flow rate for the last 7 d in the parlor before enrollment and the exposure period (retained for outcome: mean flow rate and peak flow rate), and the average milking duration for the last 7 d in the parlor before enrollment and the exposure period (retained for outcome: milking duration). Factor score from the PCA was also tested as a potential predictor in the model and retained when significant. An interaction of treatment and factor score was also tested; if an interaction of treatment and factor score was detected ($P \leq 0.05$), the data were analyzed by factor score within treatment.

The effects of treatment on the first day of having a voluntary visit to the AMS and first day of having a milking event, relative to first day of milking in the AMS (d 0), were analyzed in a linear regression model using the MIXED procedure of SAS, where the model included the fixed effects of treatment and the random effect of pair. Only cows that achieved their first voluntary visit (CON: n = 13; EXP: n = 15) or milking (CON: n = 12; EXP: n = 14) were included in the analysis. Degrees of freedom for fixed effects were estimated using the Kenward-Roger option in the MODEL statement. Factor score from the PCA was also tested as a potential predictor in the model and retained when significant. An interaction of treatment and factor score was also tested; if an interaction of treatment and factor score was detected ($P \leq 0.05$), the data were analyzed by factor score within treatment.

The effects of treatment on milking variables (milkings, concentrate delivered DM, box time, milking duration, attachment duration), milk yield, rumination, physical activity, feeding behavior, time spent around the AMS, social behaviors (pushing, head butting, fighting), ease of entry score (d 2–20), and time spent in the fetch pen during the 21-d period in the AMS were analyzed in linear regression models, using the MIXED procedure of SAS, with repeated measures. The model included the

fixed effects of treatment and day, and the random effect of pair. Several covariates were tested and retained in the model when significant ($P \leq 0.05$); these included: DIM at first AMS milking (retained for outcome: milkings), the average milk yield for the last 7 d in the parlor before enrollment and the exposure period (retained for outcomes: yield), and the average rumination time for the last 7 d before enrollment and the exposure period (retained for outcomes: rumination). To model repeated measures across time using day as the repeated measure, with cow within pair as the subject of the repeated statement, the variance-covariance matrix structure used was autoregressive or compound symmetry, chosen on the basis of best fit according to Schwarz's Bayesian information criterion. Degrees of freedom for fixed effects were estimated using the Kenward-Roger option in the MODEL statement. When treatment by day interactions were detected ($P \leq 0.05$), treatment comparisons were then performed within day, using a Tukey-Kramer correction for multiple comparisons. Factor score was also tested as a potential predictor in the model and retained when significant. An interaction of treatment and factor score was also tested; if an interaction of treatment and factor score was detected ($P \leq 0.05$), the data were analyzed by factor score within treatment.

The effects of treatment on milking variables (visits, fetches, voluntary milkings, voluntary visits, rejections, kickoffs) and social behavior (displacements at the entrance, grooming) during the 21-d period were analyzed using the GLIMMIX procedure of SAS, with either a Poisson distribution for continuous variables (visits) or binary distribution for categorized variables (fetches, voluntary milkings, voluntary visits, rejections, kickoffs, displacements at the entrance, and grooming). These models included the fixed effects of treatment and day, and the random effect of pair. Covariates were tested and retained in the models when significant ($P \leq 0.05$); DIM at first AMS milking (retained for outcome: visits). To model repeated measures across time using day as the repeated measure within the random statement, with cow within pair as the subject, the variance-covariance matrix structure used was autoregressive or compound symmetry, chosen on the basis of best fit according to the generalized chi-squared to degrees of freedom ratio. Degrees of freedom for fixed effects were estimated using the Kenward-Roger option in the MODEL statement. When treatment by day interactions were detected ($P \leq 0.05$), treatment comparisons were then performed within day, using a Tukey-Kramer correction for multiple comparisons. Factor score was also tested as a potential predictor in the model and retained when significant. An interaction of treatment and factor score was also tested; if an interaction of treatment and factor score was detected ($P \leq 0.05$), the data were analyzed by factor score within treatment.

RESULTS

Principal Components Analysis

The PCA had a sampling adequacy of 0.69. The loadings for each factor extracted from the PCA are reported in Table 4. Factor 1 explained 43.6% of the total variance and contained high positive loadings for exploring during the human test, time spent in contact with the novel human and time spent at 1 m during the human test, and negative loadings for time spent at 2 m during the human test. Cows who scored high on Factor 1 were interpreted as being "Bold." Factor 2 explained 31.1% of the total variance and contained high positive loadings for walking during the object test, escaping, and quadrant switches across all 3 tests. Cows who scored high on Factor 2 were interpreted as being "Active."

Behavior during the 48-h exposure period

During the 48-h exposure period (d −2 and −1), EXP cows had lower milk yield compared with CON cows (31.1 vs 32.6 kg/d; SE = 0.37; treatment df = 1, 29; day df = 1, 30; $P = 0.008$). There were treatment by day interactions for both rumination and activity ($P < 0.001$). Rumination (treatment df = 1, 29; day df = 1, 30) for EXP cows on d −2 was lower for EXP cows compared with CON cows (533.2 vs 582.8 min/d; SE = 16.63; $P = 0.03$; Figure 2a), while there was a tendency for EXP cows to have higher rumination on d −1 compared with CON cows (587.5 vs 545.9 min/d; SE = 16.63; $P = 0.08$; Figure 2a). Activity (treatment df = 1, 30; day df = 1, 30) for EXP cows was higher on d −2 compared with CON cows (663.7 vs 510.6 u/d; SE = 26.30; $P < 0.001$; Figure 2b), but there was no difference detected for d −1 (591.1 vs 537.6 u/d; SE = 26.30; $P = 0.2$; Figure 2b). While the AMS was set to pass through during this period, only 1 EXP cow entered the AMS and was let out without being milked, and this only occurred once.

Influence of the exposure period on performance when milked in the AMS

During the first 7 milkings, EXP cows had better (lower) milk letdown, kicking, and ease of entry scores compared with CON cows (Table 5). There was no difference detected for pellet consumption scores across the first 7 milkings (Table 5). Box time for the first 7 milkings was also shorter for EXP cows, with both milking duration and attachment duration tending to be shorter for EXP cows compared with CON cows (Table 5). There

were treatment by session interactions for milk yield, mean milk flow, and peak milk flow across the first 7 milkings. The EXP cows had greater milk yield during the first milking session compared with CON cows (12.6 vs 7.0 kg/milking; SE = 0.71; $P < 0.001$), and had greater mean milk flow (1.05 vs 0.77 kg/min; SE = 0.04; $P < 0.001$) and peak milk flow (2.18 vs 1.80 kg/min; SE = 0.12; $P = 0.05$) during the first milking session in the AMS compared with CON. No other treatment differences were detected for outcome variables relating to the first 7 milkings (Table 5). While the AMS was set to pass through outside of the milking times during the first 7 milkings, 5 EXP cows and 2 CON cows entered the AMS and were rejected during this time.

The EXP cows had more frequent AMS visits compared with CON cows across the 21-d period (Table 6). The EXP cows also tended to have a 1.39 × lower odds of requiring daily fetching (OR = 0.72; 95% CI = 0.51–1.01; $P = 0.06$), and 2.62 × lower odds of having a kickoff event (OR = 0.38; 95% CI = 0.23–0.63; $P < 0.001$) compared with CON cows. Finally, EXP cows had greater daily milk yield compared with CON cows (Table 6). Ease of entry score was also better for EXP cows compared with CON cows across the 21-d period after introduction to the AMS (0.54 vs 1.04; SE = 0.15; $P = 0.02$). No other treatment differences were detected for milking behaviors (Table 6).

There was a treatment by day interaction for the social behavior of cows being head butted by others (Table 7). Events where cows were head butted by others around the AMS were less frequent on d 0 for EXP cows compared with CON cows (10.9 vs 25.4 events/d; SE = 0.2; $P = 0.002$). No other treatment differences were detected for time spent around the AMS and social behaviors across the 21-d period (Table 7).

Across the 21-d period of milking in the AMS, EXP cows had greater rumination compared with CON cows (Table 8, Figure 2a). Treatment by day interactions for activity, meal frequency, meal size, and feeding time were detected (Table 8). Activity was lower on d 0 for EXP cows compared with CON cows (648.9 vs 809.1 u/d; SE = 33.3; $P = 0.002$; Figure 2b). Meal frequency was lower for EXP cows on d 0 compared with CON cows (8.4 vs 12.1 meals/d; SE = 0.7; $P < 0.001$). Meal size was larger for EXP cows on d 0 compared with CON cows (2.6 vs 1.7 kg/meal; SE = 0.3; $P = 0.03$). Feed time was higher for EXP cows on d 0 compared with CON cows (176.3 vs 140.9 min/d; SE = 15.5; $P = 0.05$). No other treatment differences were detected for feeding behavior for any other days of the 21-d period (Table 8).

Table 4. Coefficients (loadings) of the eigenvectors for the factors extracted from each principal components analysis of behaviors recorded when cows (n = 32) were subjected to a combined arena test[1,2]

Behaviors	Factor 1	Factor 2
Locomotion during the object test	0.18	**0.88**
Quadrant switches across all three tests[3]	0.23	**0.90**
Escaping across the three tests	−0.30	**0.82**
Exploring during the human test[4]	**0.78**	0.08
Time spent at 2m during the novel human test	**-0.77**	0.04
Time spent at 1m during the novel human test	**0.86**	0.10
Time in contact with the novel human[4]	**0.92**	0.04
Eigenvalues	3.05	2.18
Variance Explained (%)	43.6	31.1
Interpretation[5]	Bold	Active

[1]Combined arena test consisted of successive exposure to a novel arena (10 min), novel object (10 min), and novel human (10 min).

[2]High loadings (≥|0.63|) are shown in bold, indicating behavior variables that were highly associated within each Factor.

[3]Variable was square root transformed to meet assumption of normality.

[4]Variable was ln transformed to meet assumption of normality.

[5]Personality traits were interpreted based on behaviors exhibited during the combined arena test that loaded highly (either positively or negatively) onto each factor.

Influence of cow personality on response to an exposure period

Treatment by factor score interactions were detected for the trait of bold and the variables of rejections ($P = 0.003$), voluntary visits ($P = 0.002$), and voluntary milkings ($P < 0.001$), as well as the trait of active and voluntary visits ($P = 0.02$) within EXP cows. Overall, the EXP cows had greater odds of having a rejection as compared with the CON cows (OR = 1.46; 95% CI = 1.02–2.09; $P = 0.04$). However, within the EXP cows, for every 1-unit increase in personality factor score (ranging from −1.7 to 2.1) there was a 2.5 × lesser odds of bold EXP cows having a rejection compared with less bold EXP cows (OR = 0.40; 95% CI = 0.26–0.59). Alternatively, within CON cows, there was no detected effect of factor score on the odds of having a rejection for the bold trait (OR = 1.16; 95% CI = 0.69–1.94). For voluntary visits, EXP cows had greater odds of having a voluntary visit compared with the CON cows (OR = 1.71; 95% CI = 1.18–2.46; $P = 0.004$). Within the EXP cows, however, for every 1-unit increase in factor score (ranging from −1.7 to 2.1) there was a 2.5 × lesser odds of bold EXP cows having a voluntary visit compared with less bold EXP cows (OR = 0.40; 95% CI = 0.26–0.59). Active EXP cows also had greater odds of having a voluntary visit compared with less active EXP cows (for every 1-unit increase in factor score [ranging from −1.0 to 1.1]: (OR = 4.52; 95% CI = 1.22–16.81). Alternatively, within the CON cows there was no detected association of factor score with voluntary visits for either the bold trait (OR = 1.16; 95% CI = 0.69–1.94) or active trait (OR = 0.89; 95% CI = 0.43–1.86). For voluntary milkings, overall, EXP cows had greater odds of having a voluntary milking compared with CON cows (OR = 1.54; 95% CI = 1.08–2.21; $P =$

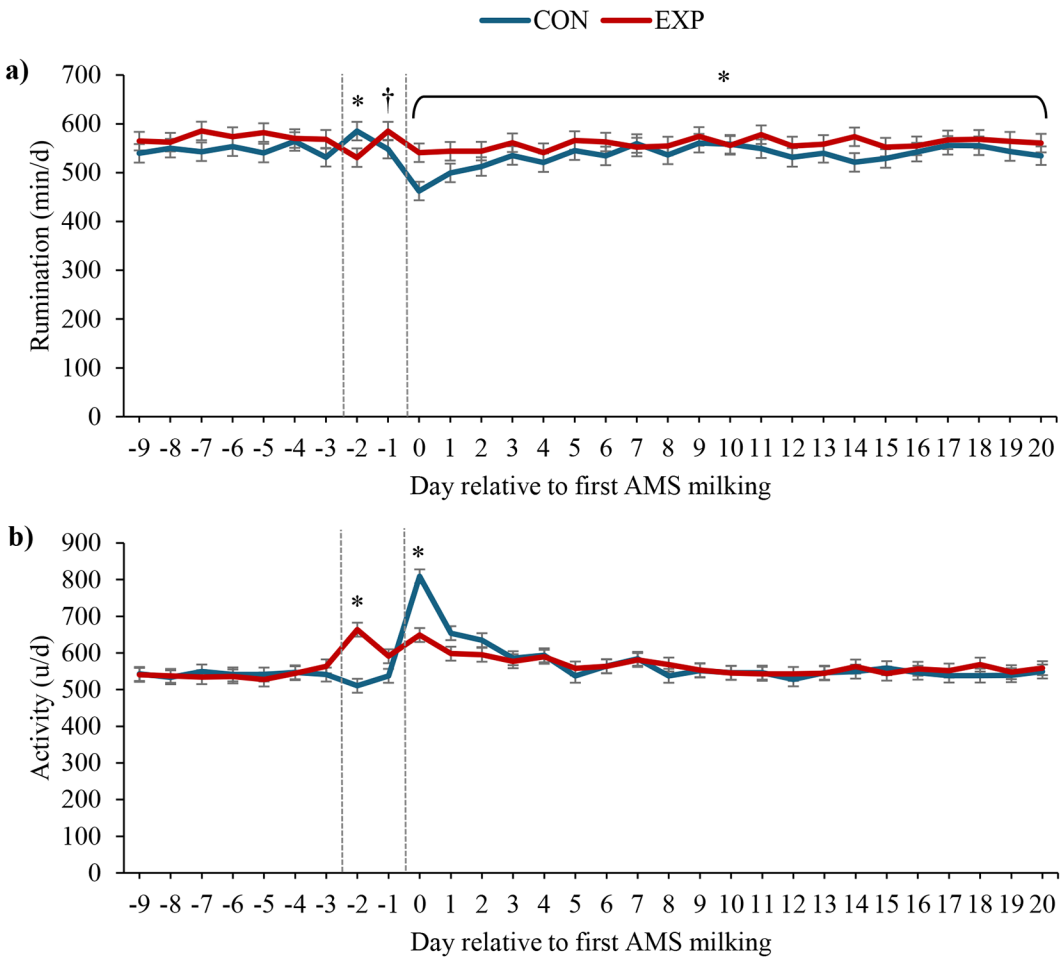


Figure 2. Mean ± SE a) rumination (min/d) and b) activity (u/d) across the study period comparing passively trained cows (EXP = experienced an exposure period in the AMS pen for 48 h before first milking in the AMS; n = 16 cows) and untrained cows (CON = no exposure period or previous exposure to the AMS; n = 16 cows). The dotted lines represent the beginning of the 48-h exposure period (starting d −2) and the beginning of the first day of milking in the AMS (starting d 0). *Indicates a significant difference ($P \leq 0.05$) and †indicates a tendency ($0.05 < P \leq 0.1$) between treatments for that period or for that day when there was a treatment x day interaction ($P < 0.001$) for that period.

0.02). However, within the EXP cows, for every 1-unit increase in factor score (ranging from −1.7 to 2.1) there was 2.94 × lesser odds of bold EXP cows having a voluntary milking compared with less bold EXP cows (OR = 0.34; 95% CI = 0.22–0.53). There was, however, no detected association in the CON cows of factor score with voluntary milkings for the bold trait (OR = 1.56; 95% CI = 0.92–2.63). No other associations between performance outcomes and factor score were detected.

DISCUSSION

Adaptation of dairy cows to an AMS can be influenced by many factors, including previous exposure, training, and individual differences in behavior. This study aimed to investigate the effects of passive training of cows on adaption using an exposure period in the AMS pen before first milking, which included the opportunity for previous exposure to the AMS and the opportunity to experience social transmission. Interestingly, only one cow out of the 16 EXP cows entered the AMS during the exposure period, so the previous exposure experienced would only apply to the external AMS environment and noises of the AMS for the majority of the EXP cows. Cows that had an exposure period in the AMS pen had better ease of entry, kicking, and milk letdown scores for the first 7 milkings compared with CON cows, indicating that these cows may have been less stressed during milking. This may be due to the fact that the EXP cows were habituated to the AMS through either its noises or external environment (Brasier et al., 2025b), as well as the AMS pen and the cows within the pen during the exposure period. This was also supported through greater milk yield, and greater mean and peak milk flow, during the first milking session

ARTICLE IN PRESS—UNCORRECTED PROOF

Table 5. Effect of an exposure period in the automated milking system (AMS) pen before first milking on milk letdown, kicking, pellet consumption, and ease of entry scores, as well as milk yield, duration, and flow rate during the first 7 milking sessions

	Treatment[1]			P value[2]		
Variable	CON	EXP	SEM	Treatment	Session[3]	Treatment × Session[3]
Milk letdown[4,5]	1.05	0.59	0.13	0.006	<0.001	0.36
Kicking[4,6]	0.74	0.23	0.07	<0.001	<0.001	0.41
Pellet consumption[4,7]	0.59	0.14	0.07	0.11	0.35	0.23
Ease of entry[4,8]	4.45	3.57	0.14	<0.001	<0.001	0.15
Milk yield (kg/milking)[4]	9.80	11.28	0.36	0.01	<0.001	<0.001
Box time (min/milking)[4]	7.41	6.61	0.03	0.02	<0.001	0.40
Milking duration (min/milking)[4]	5.50	4.94	0.20	0.06	<0.001	0.65
Attachment duration (min/milking)[4]	1.91	1.68	0.08	0.07	0.005	0.48
Mean milk flow (kg/min)[4]	0.95	1.05	0.03	0.03	<0.001	<0.001
Peak milk flow(kg/min)[4]	2.03	2.17	0.11	0.36	<0.001	0.004

[1]CON = no exposure period or previous exposure to the AMS (n = 16 cows), EXP = experienced an exposure period in the AMS pen for 48 h before first milking in the AMS (n = 16 cows).

[2]Degrees of freedom for all variables: treatment df = 1, 29; session df = 6, 180, except for behavior scores and attachment duration: treatment df = 1, 30; session df = 6, 180.

[3]Milking session: 3x/d for 2 d (d 0 and 1), plus 1 on d 2 for a total of 7 milking sessions in the AMS after introduction.

[4]Average of scores or duration collected during the first 7 milkings in the AMS.

[5]Letdown score: 0 = good, 1 = acceptable, 2 = poor, 3 = bad.

[6]Kicking score: 0 = no kicking or stepping, 1 = stepping, 2 = 2 or less kicking events, 3 = 2 or more kicking events.

[7]Pellet consumption score: 0 = ate concentrate, 1 = ate some concentrate, 2 = did not eat concentrate.

[8]Ease of entry score: 0 = went in on own, 1 = guided toward the AMS, 2 = lightly touched, 3 = pushed in, 4 = used a pinch gate, 5 = used a pinch gate and pushed for 30 s, 6 = used a pinch gate and extra people.

for EXP cows compared with CON cows. Further, the EXP cows had shorter box times, milking durations, and attachment durations across the first 7 milkings, likely related to the better kickoff scores during milking and improved milk letdown.

Undesirable milking behaviors (kickoff events) were exhibited more often by CON cows compared with EXP cows across the 21-d period. Primiparous cows have been reported to have more undesirable milking behaviors in AMS compared with multiparous cows, such as kickoffs and low milk flow (Munoz-Boettcher et al., 2025), potentially due to the novel environment. This indicates that the EXP cows adapted better to their novel environment, likely due to the habituation to the external

Table 6. Effect of an exposure period in the automated milking system (AMS) pen before first milking on milking variables of primiparous cows from d 2 to 20 during the adaptation period in the AMS

	Treatment[1]			P value[2]		
Variable	CON	EXP	SEM	Treatment	Day	Treatment × Day
Total AMS visits (#/d)[3]	5.5	6.8	0.28	0.002	<0.001	0.98
Milking frequency (#/d)[3]	2.7	2.8	0.16	0.59	<0.001	0.91
Milk yield (kg/d)[3]	32.5	33.5	0.29	0.02	0.33	0.96
Box time (min/milking)[3]	6.7	6.9	0.36	0.66	<0.001	0.92
Milking duration (min/milking)[3]	5.1	5.4	0.25	0.31	<0.001	0.93
Attachment duration (min/milking)[3]	1.6	1.5	0.01	0.18	0.08	0.54
Mean milk flow (kg/min)[3]	0.9	0.9	0.02	0.87	0.33	0.12
Peak milk flow (kg/min)[3]	1.4	1.4	0.04	0.76	0.04	0.17
First voluntary visit (d)[4]	5.7	4.5	0.89	0.32	—	—
First voluntary milking (d)[4]	6.2	6.6	1.12	0.76	—	—

[1]CON = no exposure period or previous exposure to the AMS (n = 16 cows), EXP = experienced an exposure period in the AMS pen for 48 h before first milking in the AMS (n = 16 cows).

[2]Degrees of freedom for all variables: treatment df = 1, 30; day df = 18, 540, except for milk yield: treatment df = 1, 29; day df = 18, 558, the first voluntary visit: treatment df = 1, 26, and the first voluntary milking treatment df = 1, 24.

[3]Average of data collected from d 2 to 20 of the observation period of milking in the AMS after introduction.

[4]Relative to the first day of being milked in the AMS. Only cows that achieved their first voluntary visit and milking were included in analysis.

Table 7. Effect of an exposure period in the automated milking system (AMS) pen before first milking on time spent in the fetch pen across the 21-d period, time spent around the AMS, and social behaviors of primiparous cows on d 0, 2, 6, 13, and 20 in the AMS

	Treatment[1]			P value[2]		
Variable	CON	EXP	95% CI	Treatment	Day	Treatment × Day
Time in the fetch pen (min/d)[3,4]	27.8	16.7	19.1, 25.4	0.25	<0.001	0.99
Time at the entrance (min/d)[3,5]	33.1	36.4	35.1, 46.9	0.54	<0.001	0.78
Time at the exit (min/d)[3,5]	14.6	15.6	9.8, 13.6	0.87	<0.001	0.69
Time at the crossover (min/d)[3,5]	9.0	9.4	8.5, 11.9	0.93	0.57	0.62
Total time around the AMS (min/d)[3,5]	56.7	61.5	56.1, 69.7	0.60	<0.001	0.59
Pushing (count/d)[3,5]	2.7	2.2	2.2, 3.2	0.49	0.05	0.38
Pushed (count/d)[3,5]	3.3	3.8	3.3, 4.7	0.46	0.09	0.65
Head butting (count/d)[3,5]	4.9	4.0	3.8, 5.6	0.80	0.06	0.30
Head butted (count/d)[3,5]	14.1	11.5	9.7, 12.5	0.62	<0.001	0.001
Fighting (count/d)[3,5]	4.3	3.8	2.8, 4.0	0.87	0.002	0.07

[1]CON = no exposure period or previous exposure to the AMS (n = 16 cows), EXP = experienced an exposure period in the AMS pen for 48 h before first milking in the AMS (n = 16 cows).

[2]Degrees of freedom for all variables: treatment df = 1, 30; day df = 4, 120, except for time in the fetch pen: treatment df = 1, 30; day df = 18, 540.

[3]Average of data collected from d 2 to 20 of the observation period of milking in the AMS after introduction.

[4]Back transformed.

[5]Average of data collected on d 0, 2, 6, 13, and 20 of milking in the AMS after introduction.

AMS environment and noises of the AMS, resulting in improved behavior in the AMS. Further evidence that the EXP cows adapted better to the AMS compared with the CON cows is seen in that they had lesser odds of being fetched, thus requiring less human intervention to visit the AMS, and had increased milk yield across the 21-d period. These results are similar to previous studies where cows with previous exposure to the AMS were compared with cows that did not (Brasier et al. 2025a, b). The current study, similar to a study by von Kuhlberg et al. (2021), can be defined as passive training, as cows were allowed to voluntarily explore or enter the AMS on their own before first milking in the AMS, and were not instructed to enter following a strict training protocol. This method of training is unlike the majority of previous studies, which have been focused on active training (Jago and Kerrisk, 2011; Widegren, 2014; Whitehead, 2024; Brasier et al., 2025a, b). Passive trained cows in the current study had fewer fetching events compared with non-trained cows, similar to active trained cows in previous studies (Brasier et al., 2025a, b). However, there was an increase in yield during the first milking session for EXP cows in the current study, as well as an average 3% increase in yield per day from d 2–20 of milking in the

Table 8. Effect of an exposure period in the automated milking system (AMS) pen before first milking on rumination, activity, and feeding behavior of primiparous cows across the 21-d period in the AMS

	Treatment[2]			P value[3]		
Variable[1]	CON	EXP	SEM	Treatment	Day	Treatment × Day
Rumination (min/d)	534.5	558.5	7.68	0.04	<0.001	0.21
Activity (u/d)	573.3	566.6	13.5	0.72	<0.001	0.001
AMS concentrate delivered DM (kg/d)	3.7	3.8	0.23	0.70	<0.001	0.94
PMR DMI (kg/d)	19.7	20.0	0.36	0.61	0.01	0.31
Total DMI (kg/d)	23.4	23.8	0.42	0.51	<0.001	0.41
PMR meal frequency (#/d)	8.1	7.4	0.37	0.21	<0.001	<0.001
PMR meal duration (min/meal)	46.8	50.1	3.54	0.51	0.20	0.38
PMR meal total time (min/d)	349.2	340.3	13.6	0.65	0.001	0.60
PMR meal size (kg/meal)	2.6	2.9	0.14	0.18	<0.001	0.01
PMR feeding time (min/d)	187.7	186.9	9.27	0.95	<0.001	0.02
PMR feeding rate (kg/min)	0.11	0.11	0.01	0.84	0.006	0.11

[1]Average of data collected across the 21-d observation period of milking in the AMS after introduction.

[2]CON = no exposure period or previous exposure to the AMS (n = 16 cows), EXP = experienced an exposure period in the AMS pen for 48 h before first milking in the AMS (n = 16 cows).

[3]Degrees of freedom for all variables: treatment df = 1, 30; day df = 20, 620, except for rumination and activity: treatment df = 1, 29; day df = 20, 569.

AMS for EXP cows compared with CON cows in the current study. This 3% increase in milk yield for the passive trained cows in the current study was lesser compared with the 6% (across the first 14 d in the AMS, Brasier et al., 2025a) and 7% (across the first 21 d in the AMS, Brasier et al., 2025b) increases in yield (persistent across the respective periods) observed in studies on actively trained cows. While cows were only followed for 21 d in these studies, Widegren (2010) reported greater milk yield for trained cows across the first 20 wk of lactation. Therefore, it is possible that the detected differences in milk yield may have persisted beyond the 21-d data collection period. Overall, this demonstrates that the extra time and labor required for active training may provide even greater benefit in terms of cow milking performance over that observed with passive training.

Another factor that may have contributed to the improved adaptation for the EXP cows was that they were acclimated for a 48-h period to the new pen and in that regrouped with new herd mates before exposure to milking in the AMS, as compared with the control cows that faced all those challenges at once. This is supported by the observed behavior patterns during the exposure period and first day of milking in the AMS. Regrouping results in more negative social behaviors (von Keyserlingk et al., 2008; Schirmann et al., 2011; Padua et al., 2022), which can include head-butting. The EXP cows in the current study experienced fewer head-butting events compared with CON cows on the first day of milking, as CON cows were new to the group, and were likely experiencing regrouping effects. Regrouping also results in lower rumination time (Schirmann et al., 2011; Padua et al., 2022; Marumo et al. 2024), which was not detected for CON cows on d 0 after starting milking in the AMS. However, EXP cows did have lower rumination on d −2, as compared with CON cows on d −2, meaning EXP cows likely experienced the negative effects of regrouping during the exposure period, instead of the first d of milking in the AMS. During the exposure period, EXP cows had higher activity on d −2, and lower activity compared with CON on d 0, suggesting that the EXP cows already experienced the social stress of regrouping during the exposure period. Regrouping typically results in reduced lying time (von Keyserlingk et al., 2008; Schirmann et al., 2011; Padua et al., 2022) and higher activity on the first day (Marumo et al. 2024), which aligns with the behavior differences relating to regrouping in the current study. Brasier et al. (2025b) focused on training heifers pre calving to the AMS and demonstrated that activity was higher for non-trained cows compared with trained cows on only the first day in the AMS pen. This may demonstrate that the familiarity with the AMS on its own has a similar effect on reducing stress compared with familiarity with the pen and pen mates. Therefore, this suggests that both previous exposure to the AMS and previous exposure to the pen and other cows in the pen could reduce the additive stress of introduction to the AMS and regrouping. Thus, the goal of passive training would be to isolate the stressors a cow faces during adaptation to the AMS. For example, if cows first experience regrouping stress during an acclimation period, then they may be able to better handle the introduction to the AMS compared with cows that experience these stressors at a single time point.

Reduced milk yield also typically occurs on the day of regrouping (von Keyserlingk et al., 2008; Smid et al., 2019; Marumo et al. 2024), which could explain why EXP cows had a lower milk yield in the parlor during the exposure period. This difference in yield is likely also driven by feed differences, as the EXP cows were eating the PMR with no AMS concentrate during the 48-h exposure period, while the CON cows were eating the TMR. These feed differences may have resulted in lesser energy intake for EXP cows compared with CON cows during the exposure period. Finally, regrouping also alters feeding behavior, such as cows having a lower DMI and a higher feeding rate on the day of regrouping (Schirmann et al., 2011). Altered feeding behavior for CON cows was only detected on d 0, which may reflect regrouping stress those cows experienced on that day. However, the stress of being milked by the AMS at the same time as regrouping for the CON cows may have exacerbated the effects on their behavior on d 0 and their performance across the 21-d period compared with EXP cows.

In addition to being habituated to the AMS environment, the AMS pen and conspecifics in the AMS pen, the improved adaptation of EXP cows to the AMS could also be attributed to social transmission, or using information they obtained from others to enhance individual learning during the exposure period (Brasier et al., 2025a). This is due to the fact that EXP cows had the opportunity to observe adapted cows utilizing the AMS during the 48-h exposure period. Bunching behavior around the AMS at peak AMS visit times (Solano et al., 2022), may have motivated EXP cows to investigate or stand around the AMS during the exposure period due to social facilitation, as they may have observed the adapted cows not being afraid of the AMS (Nawroth and Rørvang, 2022; Stenfelt et al., 2022b). EXP cows were able to observe previously adapted cows putting their head in the feed trough within the AMS (suggested to be a stimulus; Bailey et al., 2000) and hearing the concentrate drop into the feed trough during the exposure period while standing near the AMS, which may have motivated them to explore the entrance or exit of the AMS (Brasier et al., 2025a,b). However, there were no detected treatment differences for time spent at the AMS in the current study. Due to our study design, it cannot be concluded that the

ARTICLE IN PRESS—UNCORRECTED PROOF

cause of the improved adaptation for EXP cows compared with CON cows is due to social transmission alone or that EXP cows experienced social transmission. Future studies should be focused on what aspect plays the most important factor in improving adaptation, by isolating for social transmission or previous exposure. Specifically, future studies should aim to determine whether exposure to the noises while outside of the AMS, watching the AMS without other cows around, or watching the AMS that is being utilized by other cows is the most important aspect for improving adaptation.

The bolder EXP cows had lesser odds of having a daily rejection, voluntary visit, and voluntary milking. This shows that bolder EXP cows performed worse compared with less bold, or more fearful, EXP cows, as they needed more human intervention to visit and milk in the AMS. This is similar to a previous study on the association between active training and personality, where bold-vocal cows performed worse in terms of adapting to the AMS compared with less bold-vocal cows; however, this was only for cows with no previous exposure to the AMS (Brasier et al., 2026). Bolder cows are less flexible in terms of both coping style and learning ability (Finkemeier et al., 2022), which may explain why the bolder EXP cows did not perform as well in terms of adapting to the AMS compared with less bold EXP cows.

The more active EXP cows had greater odds of having a voluntary visit to the AMS. This shows that active EXP cows performed better compared with less active EXP cows, as they were able to better utilize the AMS on their own. While there are conflicting results in the literature on whether active cows adapt better to the AMS compared with less active cows (Schwanke et al., 2025; Brasier et al., 2026), the current results are similar to a previous study by Brasier et al. (2023), who demonstrated that non-trained, more active cows have more visits to the AMS after introduction and adapted better overall compared with less active cows.

Brasier et al. (2025a) demonstrated that cows trained to AMS using simpler methods (only exposure to the AMS environment) are more influenced by their personality in terms of how well they adapt. Alternatively, cows trained using more complicated methods (exposure to the AMS environment along with exposure to the robotic arm, noises, and AMS concentrate) are less influenced by their personality in terms of how well they adapt to the AMS (Brasier et al., 2025a; 2026). The method of previous exposure utilized in the current study was simpler, as it involved passive training, therefore adding to the notion that personality has a greater influence on the performance of cows trained using simpler or passive training methods. It was hypothesized that sociability would influence adaptation in the current study due to its relation to social learning (Nogues et al., 2023), and potentially social transmission; however, the social trait was not identified in the current study population of cows. Nogues et al. (2023) did note that fearful cows may use social information more readily than bolder animals, which may explain why the bolder EXP cows did not adapt as well compared with less bold or fearful EXP cows.

It is important to note that there were no associations detected between personality of the CON cows and outcomes relating to adaptation in the current study. This does not follow previous studies, as untrained cows have previously been demonstrated to be more influenced by their personality in terms of how well they adapt to the AMS (Brasier et al., 2023; 2025a; 2026). This discrepancy may be due to the fact that, in the present study, all of the CON cows consistently adapted to the AMS much worse than the EXP cows. Further, in previous studies focused on personality and its relation to adaptation to the AMS, fresh or early lactation cows have commonly been used (between 0 and 100 DIM: Brasier et al., 2023; 2025a; 2026; Johansen et al., 2025). Limited studies have been focused on mid-lactation cows (100–200 DIM); in such a study Schwanke et al. (2024b) detected no association of personality and outcomes relating to AMS adaptation. Therefore, it could be that cows in mid to late lactation may have a harder time adapting to the AMS (motivation lower; Jago and Kerrisk, 2011) regardless of personality.

In addition to some of the challenges already discussed within the interpretation of the results, further limitations for this study include: if cows truly experienced social transmission, the influence of the human-animal relationship on personality testing results, blinding to treatment, the lasting effects of regrouping, validity of the behavior scores during the first 7 milkings, and quantifying social interactions. As the EXP cows had the opportunity to explore the AMS as well as potentially experience social transmission while in the AMS pen during the same exposure period, our study design could not isolate social transmission from habituation as the cause of their improved adaptation to the AMS. Thus, it cannot be confirmed that the EXP cows truly experienced social transmission. Another limitation is that we cannot truly conclude the individual responses to the human reactivity test are due to personality alone or also due to the human-animal relationship. While the human reactivity test is standard practice for personality testing for dairy cows, the response to the novel human test may reflect the human animal-relationship (Rousing et al., 2006). Personality is shaped by both inherited aspects and individual experiences of an animal (Finkemier et al., 2018) and will likely influence the human-animal relationship. However, as the human is novel in a human reactivity test, this test should be more fear inducing and poten-

ARTICLE IN PRESS—UNCORRECTED PROOF

tially more reflective of personality over the human-animal relationship. In terms of blinding, farm staff were not blinded to treatment, as they were the ones milking the cows in the parlor during the exposure period. Bias was limited as researchers helped sort out the EXP cows for milking, and the farm staff only fetched cows when notified by the AMS during milking times as instructed. Another limitation is the possibility of lasting effects of regrouping stress. Regrouping stress may have lasted longer than 24 h and further influenced AMS adaptation. However, behavior differences between treatments that were likely caused by regrouping were only detected on the first day each cow from each treatment were in the AMS pen (d −2 and 0, respectively for EXP and CON cows). In terms of the validity of the behavior scores for the first 7 milkings, scores were adapted from previous studies, and researchers have compared the behaviors themselves to physiological indictors of stress (Hopster et al., 2002; Weiss et al., 2004, Wenzel et al., 2003; Jacobs and Siegford, 2012b). However, the scales of those scores have not been validated and should be interpreted with caution. Also, it was not possible to capture all social behaviors within the AMS pen as the camera view was limited to the AMS area, and some cows quickly moved away from other cows with no contact, making it hard to quantify these social interactions.

CONCLUSIONS

The results of this study indicate that passive training of cows through a 48-h exposure period in the AMS pen before first milking can improve adaptation and milk production when milked in the AMS. Specifically, cows that experienced an exposure period had lesser odds of having a fetching event, had more visits to the AMS, and had greater milk yield compared with cows that did not experience an exposure period. Personality influenced adaptation for cows that experienced passive training, where both bolder cows and less active cows had lesser odds of having a voluntary visits compared with less bold cows and more active cows. Overall, the results demonstrate that passive training of cows to the AMS, which requires less labor than active training methods, can improve adaptation and subsequent performance in the AMS.

NOTES

This project was financially supported by the Gustaf de Laval Fund 2025 (DeLaval International AB; Tumba, Sweden), a Natural Sciences and Engineering Research Council of Canada (Ottawa, ON, Canada) Discovery Grant, and the Canada First Research Excellence Fund (Ottawa, ON, Canada). This project also received support from the Ontario Agri-Food Innovation Alliance Research Program of the University of Guelph and the Ontario Ministry of Agriculture, Food, and Agribusiness (Guelph, ON, Canada). The authors thank the staff of the University of Guelph, Elora Research Station–Ontario Dairy Research Centre (Elora, ON, Canada). Special thanks to Hannah Koskamp (Wageningen University & Research, Wageningen, Netherlands), and to Zheqi (Cassie) Tao, Evelyn Lounsbury, Mara Espinola, Michele Brower, Emmeline te Bokkel, Lydia Conrad, Madeline Mclennan (University of Guelph, Guelph, ON, Canada), and Marion Voisot (AgroSup Dijon, France) for their help with cow handling, personality testing, or video scoring. The use of cows and all experimental procedures were in compliance with the guidelines of the Canadian Council on Animal Care (2009) and were approved by the University of Guelph Animal Care Committee (Animal Use Protocol #4903). The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: AMS = automated milking system; CON = control treatment; EXP = exposure period treatment; PMR = partial mixed ration; PCA = principal component analysis, OR = odds ratio.

REFERENCES

Bailey, D. W., L. D. Howery, and D. L. Boss. 2000. Effects of social facilitation for locating feeding sites by cattle in an eight-arm radial maze. Appl. Anim. Behav. Sci. 68:93–105. https://doi.org/10.1016/S0168-1591(00)00091-5.

Brasier, J. E., D. B. Haley, R. Bergeron, and T. J. DeVries. 2025b. Effect of training dairy heifers to an automated milking system before parturition on their adaptation and performance. J. Dairy Sci. 108:11418–11431. https://doi.org/10.3168/jds.2025-26883.

Brasier, J. E., D. B. Haley, R. Bergeron, and T. J. DeVries. 2026. Association between personality traits of primiparous dairy cows and their response to prepartum training and subsequent adaptation to an automated milking system. J. Dairy Sci. 109:5705–5716. https://doi.org/10.3168/jds.2025-27971.

Brasier, J. E., A. J. Schwanke, and T. J. DeVries. 2023. Effect of dairy cow personality traits on their adaptation to an automated milking system following parturition. J. Dairy Sci. 106:7191–7202. https://doi.org/10.3168/jds.2022-23176.

Brasier, J. E., A. J. Schwanke, D. B. Haley, R. Bergeron, and T. J. DeVries. 2025a. Effect of different training methods of dairy cows and their personality traits in relation to the automated milking system. J. Dairy Sci. 108:885–899. https://doi.org/10.3168/jds.2024-25420.

Budaev, S. V. 2010. Using principal components and factor analysis in animal behaviour research: Caveats and guidelines. Ethology 116:472–480. https://doi.org/10.1111/j.1439-0310.2010.01758.x.

Canadian Council on Animal Care (CCAC). 2009. The Care and Use of Farm Animals in Research, Teaching and Testing. CCAC.

Chapinal, N., D. M. Veira, D. M. Weary, and M. A. G. von Keyserlingk. 2007. Technical note: Validation of a system for monitoring individual feeding and drinking behavior and intake in group housed dairy cows. J. Dairy Sci. 990:5732–5736. https://doi.org/10.3168/jds.2007-0331.

Costa, J. H. C., W. G. Costa, D. M. Weary, L. C. P. Machado Filho, and M. A. G. von Keyserlingk. 2016. Dairy heifers benefit from the presence of an experienced companion when learning how to graze. J. Dairy Sci. 99:562–568. https://doi.org/10.3168/jds.2015-9387.

DeVries, T. J., M. A. G. von Keyserlingk, D. M. Weary, and K. A. Beauchemin. 2003. Measuring the feeding behavior of lactating dairy cows in early to peak lactation. J. Dairy Sci. 86:3354–3361. https://doi.org/10.3168/jds.S0022-0302(03)73938-1.

Finkemeier, M.-A., A. Krause, A. Tuchscherer, B. Puppe, and J. Langbein. 2022. Personality traits affect learning performance in dwarf goats (Capra hircus). Front. Vet. Sci. 9:916459. https://doi.org/10.3389/fvets.2022.916459.

Finkemeier, M.-A., J. Langbein, and B. Puppe. 2018. Personality research in mammalian farm animals: concepts, measures, and relationships to welfare. Front. Vet. Sci. 5:131. https://doi.org/10.3389/fvets.2018.00131.

Foris, B., M. Zebunke, J. Langbein, and N. Melzer. 2019. Comprehensive analysis of affiliative and agonistic social networks in lactating dairy cattle groups. Appl. Anim. Behav. Sci. 210:60–67. https://doi.org/10.1016/j.applanim.2018.10.016.

Galef, B. G. 1988. Imitation in animals: history, definition, and interpretation of data from the Psychological Laboratory. Pages 3–28 in Social Learning: Psychological and Biological Perspectives. T. R. Zentall and B. G. Galef, ed. Lawrence Erlbaum Associates.

Hopster, H., R. M. Bruckmaier, J. T. N. Van der Werf, S. M. Korte, J. Macuhova, G. Korte-Bouws, and C. G. van Reenen. 2002. Stress responses during milking; comparing conventional and automatic milking in primiparous dairy cows. J. Dairy Sci. 85:3206–3216. https://doi.org/10.3168/jds.S0022-0302(02)74409-3.

Jacobs, J. A., and J. M. Siegford. 2012a. Invited review: The impact of automatic milking systems on dairy cow management, behaviour, health, and welfare. J. Dairy Sci. 95:2227–2247. https://doi.org/10.3168/jds.2011-4943.

Jacobs, J. A., and J. M. Siegford. 2012b. Lactating dairy cows adapt quickly to being milked by an automatic milking system. J. Dairy Sci. 95:1575–1584. https://doi.org/10.3168/jds.2011-4710.

Jago, J., and K. Kerrisk. 2011. Training methods for introducing cows to a pasture-based automatic milking system. Appl. Anim. Behav. Sci. 131:79–85. https://doi.org/10.1016/j.applanim.2011.02.002.

Keshavarzi, H., C. Lee, J. M. Lea, and D. L. M. Campbell. 2020. Virtual fence responses are socially facilitated in beef cattle. Front. Vet. Sci. 7:543158. https://doi.org/10.3389/fvets.2020.543158.

MacDonald, P. D. M., and P. E. J. Green. 1988. User's Guide to Program MIX: An interactive program for fitting mixtures of distributions. Release 2.3, January 1988. Ichthus Data Systems, Hamilton, ON, Canada.

Martin, P., and M. Bateson. 2021. Measuring Behaviour. An Introductory Guide. 4th ed. Cambridge University Press, Cambridge, UK.

Marumo, J. L., D. Lusseau, J. R. Speakman, M. Mackie, A. Y. Byar, W. Cartwright, and C. Hambly. 2024. Behavioural variability, physical activity, rumination time, and milk characteristics of dairy cattle in response to regrouping. Animal 18:101094. https://doi.org/10.1016/j.animal.2024.101094.

Merenda, V. R., C. C. Figueiredo, T. D. González, and R. C. Chebel. 2020. Technical note: Validation of a system for monitoring individual behavior of Holstein cows. J. Dairy Sci. 103:7425–7430. https://doi.org/10.3168/jds.2019-17922.

Morris, T. R. 1999. Experimental Design and Analysis in Animal Sciences. CABI Publishing.

Munksgaard, L., A. M. DePassillé, J. Rushen, M. S. Herskin, and A. M. Kristensen. 2001. Dairy cows' fear of people: social learning, milk yield and behaviour at milking. Appl. Anim. Behav. Sci. 73:15–26. https://doi.org/10.1016/S0168-1591(01)00119-8.

Munoz-Boettcher, P., J. Velez, N. Rodriguez, I. C. Klaas, and P. Pinedo. Munoz-Boettcher. 2025. Milking behavior and performance of primiparous and multiparous Holstein, Jersey, and Holstein × Jersey crossbred cows in a batch milking system with automatic milking units. J. Dairy Sci. 108:4248–4262. https://doi.org/10.3168/jds.2024-26078.

Nawroth, C., and M. V. Rørvang. 2022. Opportunities (and challenges) in dairy cattle cognition research: A key area needed to design future high welfare housing system. Appl. Anim. Behav. Sci. 255:105727. https://doi.org/10.1016/j.applanim.2022.105727.

Nogues, E. D. M. W., and M. A. G. von Keyserlingk. 2023. Sociability, fearfulness, and coping style: Impacts on individual variation in the social behavior of dairy cattle. J. Dairy Sci. 106:9568–9575. https://doi.org/10.3168/jds.2023-23553.

Padua, F. H., K. M. Dancy, R. Bergeron, and T. J. DeVries. 2022. Impact of a dietary citrus extract on the behavior and production of lactating dairy cows following regrouping: A preliminary study. Front. Vet. Sci. 8:773399. https://doi.org/10.3389/fvets.2021.773399.

Ralphs, M. H., D. Graham, and J. F. James. 1994. Social facilitation influences cattle to graze locoweed. J. Range Manage. 47:123–126. https://doi.org/10.2307/4002819.

Ren, K., G. Bernes, M. Hetta, and J. Karlsson. 2021. Tracking and analysing social interactions in dairy cattle with real-time locating system and machine learning. J. Systems Archit. 116:102139. https://doi.org/10.1016/j.sysarc.2021.102139.

Rodenburg, J. 2017. Robotic milking: Technology, farm design, and effects on work flow. J. Dairy Sci. 100:7729–7738. https://doi.org/10.3168/jds.2016-11715.

Rørvang, M. V., J. W. Christensen, J. Ladewig, and A. McLean. 2018. Social learning in horses—Fact or fiction? Front. Vet. Sci. 5:212. https://doi.org/10.3389/fvets.2018.00212.

Rousing, T., and F. Wemelsfelder. 2006. Qualitative assessment of social behaviour of dairy cows housed in loose housing systems. Appl. Anim. Behav. Sci. 101:40–53. https://doi.org/10.1016/j.applanim.2005.12.009.

SAS Institute Inc. 2013. SAS version 9.4. SAS Institute Inc.

Schirmann, K., N. Chapinal, D. M. Weary, W. Heuwieser, and M. A. G. von Keyserlingk. 2011. Short-term effects of regrouping on behavior of prepartum dairy cows. J. Dairy Sci. 94:2312–2319. https://doi.org/10.3168/jds.2010-3639.

Schwanke, A. J., J. E. Brasier, G. B. Penner, R. Bergeron, and T. J. DeVries. 2025. Influence of dairy cow personality traits on response to extended milking intervals and removal of supplemental concentrate in a free-traffic automated milking system. J. Dairy Sci. 108:2856–2874. https://doi.org/10.3168/jds.2024-25787.

Schwanke, A. J., K. M. Dancy, H. W. Neave, G. B. Penner, R. Bergeron, and T. J. DeVries. 2022. Effects of concentrate allowance and individual dairy cow personality traits on behavior and production of dairy cows milked in a free-traffic automated milking system. J. Dairy Sci. 105:6290–6306. https://doi.org/10.3168/jds.2021-21657.

Schwanke, A. J., H. W. Neave, G. B. Penner, R. Bergeron, and T. J. DeVries. 2024a. Flexible feeding: Dairy cow personality affects changes in feeding behavior and milk production under feed competition conditions. J. Dairy Sci. 107:2465–2482. https://doi.org/10.3168/jds.2023-24063.

Schwanke, A. J., H. W. Neave, G. B. Penner, R. Bergeron, and T. J. DeVries. 2024b. Effect of dairy cow personality traits and concentrate allowance on their response to training and adaptation to an automated milking system. J. Dairy Sci. 107:11446–11462. https://doi.org/10.3168/jds.2024-25119.

Smid, A.-M. C., D. M. Weary, E. A. M. Bokkers, and M. A. G. von Keyserlingk. 2019. Short communication: The effects of regrouping in relation to fresh feed delivery in lactating Holstein cows. J. Dairy Sci. 102:6545–6550. https://doi.org/10.3168/jds.2018-16232.

Stenfelt, J., J. Yngvesson, H. J. Blokhuis, and M. V. Rørvang. 2022a. Dairy cows did not rely on social learning mechanisms when solving a spatial detour task. Front. Vet. Sci. 9:956559. https://doi.org/10.3389/fvets.2022.956559.

Stenfelt, J., J. Yngvesson, and M. V. Rørvang. 2022b. A calm companion lowers fear in groups of dairy cows. J. Dairy Sci. 105:6923–6935. https://doi.org/10.3168/jds.2022-21796.

Velázquez-Martínez, M., S. López-Ortiz, O. Hernández-Mendo, P. Díaz-Rivera, S. Pérez Elizalde, and J. Gallegos-Sánchez. 2010. Foraging behavior of heifers with or without social models in an unfamiliar site containing high plant diversity. Livest. Sci. 131:73–82. https://doi.org/10.1016/j.livsci.2010.03.001.

von Keyserlingk, M. A. G., D. Olenick, and D. M. Weary. 2008. Acute behavioral effects regrouping dairy cows. J. Dairy Sci. 91:1011–1016. https://doi.org/10.3168/jds.2007-0532.

von Kuhlberg, M. K., M. Wensch-Dorendorf, J. Gottschalk, T. Wagner, N. Herrmann, and A. Einspanier. 2021. The effects of a training program using a phantom to accustom heifers to the automatic milking

ARTICLE IN PRESS—UNCORRECTED PROOF

system. J. Dairy Sci. 104:928–936. https://doi.org/10.3168/jds.2020-18715.
Wechsler, B., and S. E. G. Lea. 2007. Adaptation by learning: Its significance for farm animal husbandry. Appl. Anim. Behav. Sci. 108:197–214. https://doi.org/10.1016/j.applanim.2007.03.012.
Weiss, D., S. Helmreich, E. Möstl, A. Dzidic, and R. M. Bruckmaier. 2004. Coping capacity of dairy cows during the change from conventional to automatic milking. J. Anim. Sci. 82:563–570. https://doi.org/10.2527/2004.822563x.
Wenzel, C., S. Schönreiter-Fischer, and J. Unshelm. 2003. Studies on step-kick behaviour and stress of cows during milking in an automatic milking system. Livest. Prod. Sci. 83:237–246. https://doi.org/10.1016/S0301-6226(03)00109-X.
Whitehead, L. 2024. Impact of pre-training on dairy cattle automatic milking system acclimation. MSC Thesis. Dept. of Animal Science, University of Tennessee, Knoxville, Tennessee.
Widegren, S. 2014. Introduction of heifers to an automatic milking system. BS Thesis. Dept. of Animal Nutrition and Management, Swedish Univ. of Agricultural Sciences, Uppsala, Sweden.

ORCIDS

J. E. Brasier, https://orcid.org/0009-0005-8750-3245
R. Bergeron, https://orcid.org/0000-0001-7757-2531
D. B. Haley, https://orcid.org/0000-0003-3502-120X
T. J. DeVries https://orcid.org/0000-0001-9364-2456